

ERODE SENGUNTHAR ENGINEERING COLLEGE · IDEATHON 2026

ThinkMark

Let AI Think. Let Teachers Teach.

India's first AI-powered, IoT-enabled exam answer sheet evaluator — built to transform how teachers grade and students learn.

Created by [Madhan.TR](#)

THE PROBLEM

Teachers Are Drowning in Paperwork

The Reality Today

- Teachers manually correct **500–1,000 papers** per exam — taking 3 to 7 exhausting days
- Human fatigue causes **inconsistent, unfair marking**
- Spelling mistakes punish students who understand the concept
- No existing tool supports **Tamil handwritten answers**
- Enterprise AI tools cost **₹4.5 lakh/year** — unaffordable for schools

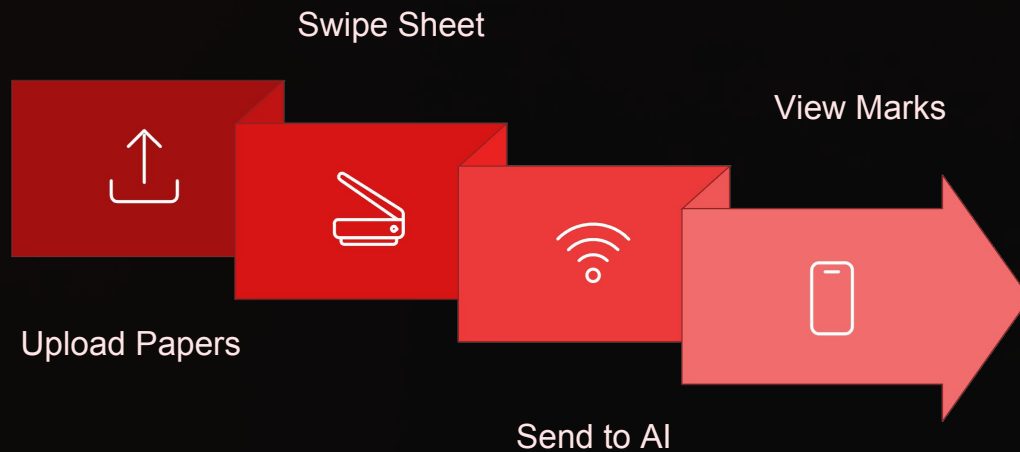
The Scale of Impact

1.2 crore students write Tamil Nadu board exams every year. Every single paper is corrected manually by an exhausted teacher.

OUR SOLUTION

What Is ThinkMark?

A **T-shaped handheld scanner** — swipe an answer sheet, and AI evaluates it in seconds. No app needed.



Supports theory answers, math equations, graphs, chemical equations, Tamil, English — all subjects.

ThinkMark Web Dashboard

What Teachers See After Every Scan

(i) Student: Arun Kumar | Roll: 2024CS042 | Mathematics

- Q1 (16 marks) — **13/16**
- Q2 (16 marks) — **14/16**
- Q3 (18 marks) — **15/18**
- Total: **42/50 — PASS** | Class Rank: 2/60

Dashboard Features

- Works on **any phone browser** — zero setup
- Admin monitoring — see all scans live
- JWT security + duplicate scan blocking
- Green = Pass, Red = Fail — instant visual result



INSTANT RESULTS



LIVE ADMIN MONITORING



SECURITY LAYER



VISUAL PASS/FAIL

Tech Stack & What Makes Us Unique

Hardware

- Arduino Uno + ESP8266 WiFi + Camera module
- 18650 battery + 3D-printed T-shape body

Software

- Node.js backend + Gemini 2.0 Flash AI
- Sarvam AI (Tamil) + JWT security
- Netlify + Render.com

Why ThinkMark Stands Alone

Feature	Competitors	ThinkMark
Physical IoT Hardware	Nobody	✓ Only one
Tamil Language Support	Nobody	✓ Sarvam AI
Graph Evaluation	Very rare	✓ Yes
Price	₹4.5L/year	✓ ₹3,999
Browser Access	Partial	✓ Zero setup

Challenges & Our Solutions

Handwriting Variability

Gemini Vision AI trained on millions of handwriting samples.

தமிழ் Tamil OCR Accuracy

Sarvam AI — built specifically for Indian language handwriting.

சுருள் Graph & Diagram Recognition

Gemini Vision API with shape detection capabilities.

தாமத Internet Dependency

2–3 second response on 4G + local caching for speed.

Security & Tampering

JWT auth + backend-only mark calculation + rate limiting + duplicate scan blocking.

Social & Environmental Impact

40hrs

Saved Per Teacher

Per exam season freed from manual correction

100x

Cheaper

Than existing enterprise solutions

37K+

Schools Can Benefit

Tamil Nadu government schools eligible

₹3,999

Device Cost

Affordable for every school

➤ Social Impact

- Saves 3–7 days of teacher correction time per exam
- Fair, unbiased marking — no fatigue-based errors
- First tool to support Tamil language exam grading in India
- Students get results faster — less anxiety

Environmental Impact

- Fully digital reports — reduces paper usage
- Battery operated — low power consumption
- 3D-printed body — recyclable materials

PROTOTYPE

ThinkMark v0.1 — Current Status

✓ Already Working

- Full website live at thinkmark.nellify.app
- Admin dashboard with security monitoring
- Scan simulation with pass/fail results
- JWT authentication and security system
- Arduino WiFi web server code written
- GitHub repository published (MIT license)

🔧 In Progress

- Camera module integration
- Real Gemini AI backend connection
- Tamil OCR via Sarvam AI
- 3D-printed T-shape hardware fabrication
- Field testing in school environment

📘 ThinkMark is a working prototype with a live demo — the full hardware pipeline is in active development.

Market Opportunity & Revenue Model

Target Market

- Tamil Nadu Government Schools: **37,000+**
- Engineering Colleges in Tamil Nadu: **500+**
- Coaching Centres: **10,000+**

India EdTech Market 2026: **\$10.4 Billion**

Revenue Streams

- Device sale: **₹3,999** per unit
- Monthly SaaS: **₹499/month**
- Per paper: **₹2/paper** for exam boards
- Enterprise license: **₹50,000/year**

Growth Roadmap

Year 1 — Pilot

Free pilot across 5 schools in Erode.
Validate product-market fit.

1

2

Year 2 — Scale

Sell to 100 schools — target **₹40 lakh** revenue.

3

Year 3 — Partnership

Tamil Nadu government partnership — target **₹5 crore** revenue.

THINKMARK

Let AI Think. Let Teachers Teach.

Every teacher in Tamil Nadu should grade 60 answer sheets in **10 minutes** — not 3 days.

🔗 Live Demo

thinkmark.netlify.app

🌊 GitHub

github.com/madhantr0/thinkmark

👤 Creator

[Madhan.TR](https://www.linkedin.com/in/madhantr)

🏢 Institution

Erode Sengunthar Engineering College · Ideathon 2026

Arduino · Node.js · Gemini AI · Sarvam AI · Netlify · JWT Security